



A New Test for Assessing the Covariate Effect in ROC Curves

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Abstract

This paper investigates the impact of covariates on ROC curves, comparing Pooled ROC, Conditional ROC and AROC models. After explaining the methodology described in the literature, we performed some simulations to see how these models perform in real-world scenarios, concluding that in less complex situations the Pooled ROC is the best model that we can choose.

1 Introduction

The **ROC** (*Receiver Operating Characteristic*) framework acts as a statistical method to quantify how effectively a diagnostic variable distinguishes between two distinct populations, it is primarily used in medicine to divide between healthy and diseased patients.

As a result, the ROC curve is the combination of sensitivity and specificity of diagnostic variables to verify the discrimination's accuracy between the groups.

Sensitivity, also called the *True Positive Rate*, is the proportion of actual diseased individuals who are correctly identified by the test:

$$\text{Sensitivity} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (1)$$

A test with low sensitivity incorrectly classifies diseased individuals as healthy, resulting in a *Type II Error* (False Negatives).

Specificity, also called the *True Negative Rate*, is the proportion of actual healthy individuals who are correctly classified as healthy by the test:

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}} \quad (2)$$

A test with low specificity incorrectly classifies healthy individuals as diseased, resulting in a *Type I Error* (False Positives).

The covariates are some characteristics of the groups, such as age or gender, that can affect the diagnostic variable. They are fundamental because they can affect our study, since including them in the ROC curve analysis can lead to three different situations:

- The performance of the ROC curve changes with the covariate.
- The covariate affects the distribution of the diagnostic markers, but not their discriminatory capability.
- The covariate does not affect the ROC curve at all.

We refer to the standard ROC curve as the **Pooled ROC** because it is obtained by combining all observations from multiple groups into a single dataset, disregarding covariate information.

1.1 Pooled ROC

We introduce some notations to fully understand the curve: Y^F and Y^G are the continuous variables in the diseased and healthy populations, X^F and X^G are the covariates associated with these populations and R_X is the intersection of the support of X^G and X^F . The **Pooled ROC** can be defined as:

$$\text{ROC}_p(p) = 1 - F(G^{-1}(1 - p)), \quad p \in (0, 1) \quad (3)$$

where $F(y) = P(Y^F \leq y)$ and $G(y) = P(Y^G \leq y)$ are the cumulative distribution functions of Y^F and Y^G , respectively, and G^{-1} is the quantile function associated to G .

1.2 Conditional ROC

Considering $x \in R_X$, the covariate with fixed value, the **Conditional ROC** is defined as:

$$\text{ROC}^x(p) = 1 - F(G^{-1}(1 - p|x)|x), \quad p \in (0, 1) \quad (4)$$

where $F(\cdot|x)$ and $G(\cdot|x)$ are the cumulative distribution functions of the diagnostic variable in the diseased and healthy population, respectively, conditioned to the value of the covariate x . This is the best choice in the first case, so when the performances of the curve change with the covariate.

1.3 AROC Curve

The **Covariate-Adjusted ROC (AROC)** curve is the curve obtained when we employ a covariate-specific threshold, this means that we choose a threshold value for the diagnostic value above which the individual can be considered healthy or diseased. The threshold is *covariate specific* because it varies according to patient characteristics; for example the optimal threshold for a young patient differs from the one for an elderly patient. It can be expressed as:

$$\text{AROC}(p) = P(Y^F > G^{-1}(1 - p|X^F)), \quad p \in (0, 1) \quad (5)$$

The AROC curve can also be expressed as an expectation of Conditional ROC curves:

$$\text{AROC}(p) = \int \text{ROC}^x(p) dF_{X^F}(x) \quad (6)$$

where $F_{X^F}(x) = P(X^F \leq x)$ is the cumulative distribution function of the covariate in the diseased population.

The AROC curve is the best when summarizing the performance of the Conditional ROC curve when sample sizes are insufficient for accurate local estimation. When we have very small samples some problems may arise but thanks to the AROC, which summarizes the overall performance by weighting the information across the entire covariate distribution, we are able to improve the study. Notably, when the covariate affects the diagnostic variables but not their discriminatory accuracy, we get:

$$\text{AROC}(p) = \text{ROC}^x(p) \quad (7)$$

1.4 How to Choose the Right Curve

First, it is fundamental to test whether the Conditional ROC curves is affected by the values of the covariate. If the curves coincide with the AROC ($\text{ROC}^x(p) = \text{AROC}(p)$), the covariate does not influence discriminatory power. However, covariates might still have an effect on the diagnostic variable's distributions.

Thus, we are interested in the test for the **null hypothesis**:

$$H_0 : \text{AROC}(p) = \text{ROC}(p), \quad p \in (0, 1) \quad (8)$$

vs. the alternative $H_1 : \text{AROC}(p) \neq \text{ROC}(p)$ for $p \in (0, 1)$.

If the null hypothesis is rejected, the AROC curve should be employed because choosing the Pooled ROC, which disregards the covariate effect, might be leading to inaccurate results. On the other hand, if H_0 is accepted, the Pooled ROC is preferred because of its simplicity, as the Pooled Roc is computationally simpler and better documented in the literature.

1.5 Visual Interpretation

Assume that we are using a finite sample study to analyze the performance of the test, considering four scenarios (A, B, C, and D), each different because of the covariate:

Scenario A: The covariate has no effects and the distributions do not change. Consequently, the Pooled ROC, Conditional ROC and AROC coincide:

$$\text{ROC}(p) = \text{ROC}^x(p) = \text{AROC}(p)$$

Scenario B: The covariate affects only one population (the distribution of the healthy population changes, while the diseased one remains constant). The Pooled ROC and AROC coincide:

$$\text{ROC}(p) = \text{AROC}(p)$$

However, the Conditional ROC curves vary: they are close to the diagonal for lower covariate values and close to maximum accuracy for higher values.

Scenario C: The covariate affects both populations and both distributions change in a parallel way. Since the discriminatory capability remains constant, the Conditional ROC and AROC coincide, but differ from the Pooled ROC:

$$\text{ROC}^x(p) = \text{AROC}(p) \neq \text{ROC}(p)$$

Scenario D: The covariate affects both populations, and the densities change in an irregular way. All three curves are different:

$$\text{ROC}(p) \neq \text{ROC}^x(p) \neq \text{AROC}(p)$$

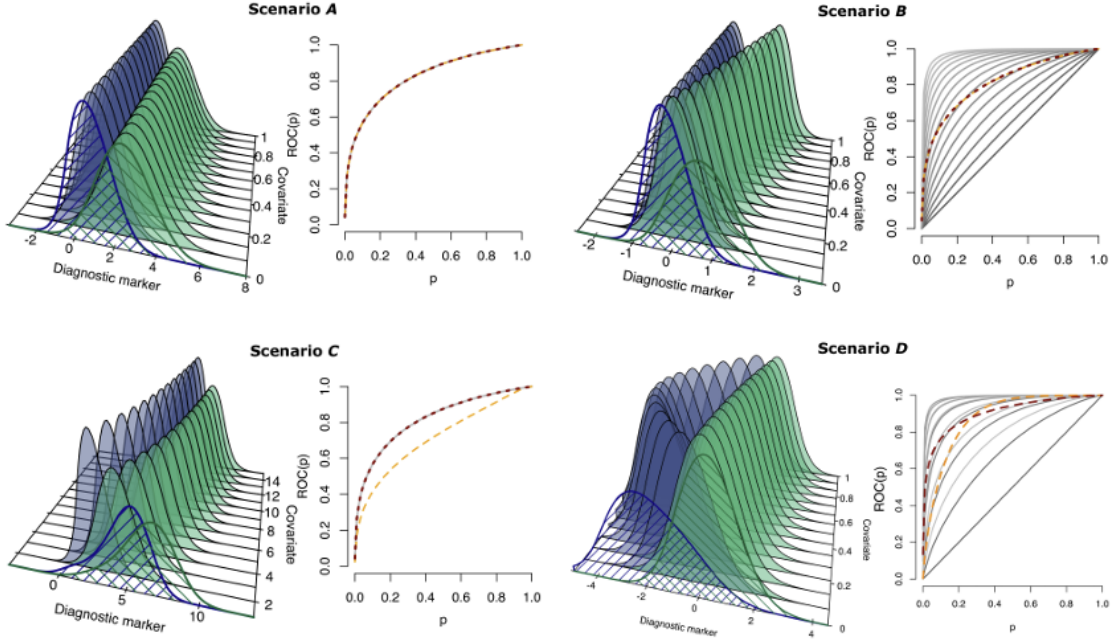


Figure 1: Four scenarios illustrating the relationship among the Pooled, Conditional, and Covariate-Adjusted ROC curves. The Pooled ROC is in orange, the Conditional ROC in gray, and the AROC in brown. The density of the diagnostic variable is represented in green for the diseased population and in blue for the healthy population.

2 Methodology

The study methodology proposes a **nonparametric test** to verify if the covariate influences the performance of the diagnostic variable or not. Firstly, we introduce two i.i.d. samples, $\{(X_i^F, Y_i^F)\}_{i=1}^{n^F}$ from the diseased population and $\{(X_i^G, Y_i^G)\}_{i=1}^{n^G}$ from the healthy population. (X^F, Y^F) and (X^G, Y^G) are assumed to be independent.

For estimating the Pooled ROC we employ the empirical estimator, obtaining:

$$\widehat{ROC}(p) = 1 - \hat{F}(\hat{G}^{-1}(1 - p)), \quad p \in (0, 1) \quad (9)$$

For estimating the AROC we use **nonparametric location-scale regression models** which capture relationship between the diagnostic marker and the covariate. Given $D \in \{F, G\}$, the notation is as follows: X^D is the covariate associated with Y^D , $\mu^D(\cdot)$ and $(\sigma^D)^2(\cdot)$ are the conditional mean and the conditional variance functions, and the error ϵ^D is independent of X^D and has cumulative distribution function H^D . Getting:

$$Y^D = \mu^D(X^D) + \sigma^D(X^D)\epsilon^D \quad (10)$$

With location-scale regression models we have the conditional quantile expressed in terms of the error distribution. At this point the AROC estimator is defined as:

$$\widehat{AROC}(p) = \frac{1}{n^F} \sum_{i=1}^{n^F} I \left(\hat{H}^G \left(\frac{Y_i^F - \hat{\mu}^G(X_i^F)}{\hat{\sigma}^G(X_i^F)} \right) > 1 - p \right) \quad (11)$$

Then we compare the estimators of the Pooled and the Covariate-adjusted ROC curves using a test statistic of the form:

$$S_\psi = \psi(\widehat{ROC}(p) - \widehat{AROC}(p)) \quad (12)$$

where ψ is a continuous function chosen to measure the distance between the curves. Then three distance functions are introduced, two based on L_1 and L_2 and the other one on Kolmogorov-Smirnov:

$$S_{\psi_{L_1}} = \int |\widehat{ROC}(p) - \widehat{AROC}(p)| dp \quad (13)$$

$$S_{\psi_{L_2}} = \int (\widehat{ROC}(p) - \widehat{AROC}(p))^2 dp \quad (14)$$

$$S_{\psi_{KS}} = \sup_{p \in (0,1)} |\widehat{ROC}(p) - \widehat{AROC}(p)| \quad (15)$$

In the first two cases, we are calculating the **area** between the two curves, whereas in the third one we are calculating the **maximum gap** between them. Under the null hypothesis they take values close to zero, while when we reject H_0 they take larger positive values. If we accept the null hypothesis $S \sim 0$, this implies that the area or the maximum distance between the curves is also very close to zero so they are identical or almost overlapping. On the other hand, if $S > 0$ they can not be identical because the area or the distance between them is not zero.

Then, we use a **bootstrap algorithm** based on the general one. Firstly, we consider:

$$T_\psi = \psi((\widehat{ROC}(p) - ROC(p)) - (\widehat{AROC}(p) - AROC(p))) \quad (16)$$

where $\psi((\widehat{ROC}(p)))$ and $\psi((\widehat{AROC}(p)))$ are the estimated curves calculated from the specific sample, and $ROC(p)$ and $AROC(p)$ are the theoretical curves which represent the ideal situation if we could observe the entire population. Under H_0 it is not possible to use S_ψ ; so, since if the null hypothesis holds the following equivalence is true $S_\psi = T_\psi$, we can employ T_ψ . At the end, in order to avoid dependency problems, we randomly split the original sample into two sets, getting $\{(X_{A,i}^F, Y_{A,i}^F)\}_{i=1}^{n_A^F}$ and $\{(X_{A,j}^G, Y_{A,j}^G)\}_{j=1}^{n_A^G}$ for the AROC curve, and $\{Y_{R,i}^F\}_{i=1}^{n_R^F}$ and $\{Y_{R,j}^G\}_{j=1}^{n_R^G}$ for the ROC curve, with:

$$n^F = n_R^F + n_A^F \quad \text{and} \quad n^G = n_R^G + n_A^G \quad (17)$$

Calculating the distribution of the statistics T_ψ under the null hypothesis is analytically too complicated, so we use a bootstrap-based procedure to find the p-values and the critical values.

Algorithm: Bootstrap Resampling Procedure

We perform B bootstrap iterations to check the null hypothesis $H_0 : ROC(p) = AROC(p)$

1. **Data Splitting:** We split the original sample into two sets: a set $\{Y_{R,i}^F\}_{i=1}^{n_R^F}$ and $\{Y_{R,i}^G\}_{i=1}^{n_R^G}$ used to construct the ROC, and a set $\{(X_{A,i}^F, Y_{A,i}^F)\}_{i=1}^{n_A^F}$ and $\{(X_{A,i}^G, Y_{A,i}^G)\}_{i=1}^{n_A^G}$ for the construction of the AROC, considering both the diseased (F) and healthy (G) populations.

2. **Resampling Step:** For each iteration $b = 1, \dots, B$:

- **For Set R (Empirical):** For $D \in \{F, G\}$, let $\{Y_{D,b,R,i}^*\}$ be an i.i.d. sample drawn from the empirical distribution of the first set of data.
- **For Set A (Residuals):** For $D \in \{F, G\}$, we draw residuals $\{\varepsilon_{D,b,A,i}^*\}$ from the empirical distribution of the residuals computed on the second set. We reconstruct the bootstrap sample using the location-scale model:

$$Y_{D,b,A,i}^* = \hat{\mu}_D(X_{D,A,i}) + \hat{\sigma}_D(X_{D,A,i})\varepsilon_{D,b,A,i}^*$$

3. **Bootstrap Estimates Construction:** For each iteration $b \in \{1, \dots, B\}$, compute the curves using the split samples:

- **Pooled ROC:** Compute $\widehat{ROC}_b^*(p)$ for $p \in (0, 1)$ using the resampled data from the ROC set:

$$\{Y_{D,b,R,i}^*\}_{i=1}^{n_D^R}, \quad D \in \{F, G\}$$

- **AROC:** Compute $\widehat{AROC}_b^*(p)$ for $p \in (0, 1)$ using the reconstructed data from the AROC set:

$$\{(X_{D,A,i}, Y_{D,b,A,i}^*)\}_{i=1}^{n_D^A}, \quad D \in \{F, G\}$$

4. **Statistic Calculation:** For each iteration $b = 1, \dots, B$, we compute the bootstrap statistic taking into account the difference between the bootstrap estimation and original estimation (we don't consider the theoretical one like in T_ψ):

$$T_{\psi,b}^* = \psi \left(\left(\widehat{ROC}_b^*(p) - \widehat{ROC}(p) \right) - \left(\widehat{AROC}_b^*(p) - \widehat{AROC}(p) \right) \right), p \in (0, 1) \quad (18)$$

5. **P-value Estimation:** With the following formula embodying the p-value definition we are able to estimate the p-value. We calculate the frequency of cases where the random sampling noise produce a discrepancy as large as the observed one:

$$p\text{-value} = \frac{1}{B} \sum_{b=1}^B I(T_{\psi,b}^* \geq T_\psi)$$

Using the indicator function, we observe that if the bootstrap statistics is smaller than the observed one, the p-value will be very small and we reject the null hypothesis. Contrarily, we can accept the null hypothesis since the identity function will be equal to 1.

3 Simulations

We now reconsider the four scenarios introduced in the visual interpretation. We assume regression errors ε^F and ε^G follow a standard normal distribution. In Scenarios A, B and C the covariate follows a uniform distribution on the unit interval for both the diseased and healthy populations, while in Scenario D it follows a uniform distribution on the interval $[1, 15]$.

We consider three different sample sizes, with the second one being unbalanced: $(n^F, n^G) = (100, 100), (250, 350), (500, 500)$. We use three different distances $(\psi_{L_1}, \psi_{L_2}, \psi_{KS})$ for the construction of the **test statistics**.

In the following graphs, the dotted horizontal lines represent the nominal levels ($\alpha = 0.025, 0.05, 0.1$), where the dots should ideally sit. The vertical colored bars represent the confidence intervals; if the dotted line passes through a bar, the test is considered well-calibrated. The axes are defined as follows: the x -axis represents the sample sizes considered and the y -axis represents the proportion of rejections.

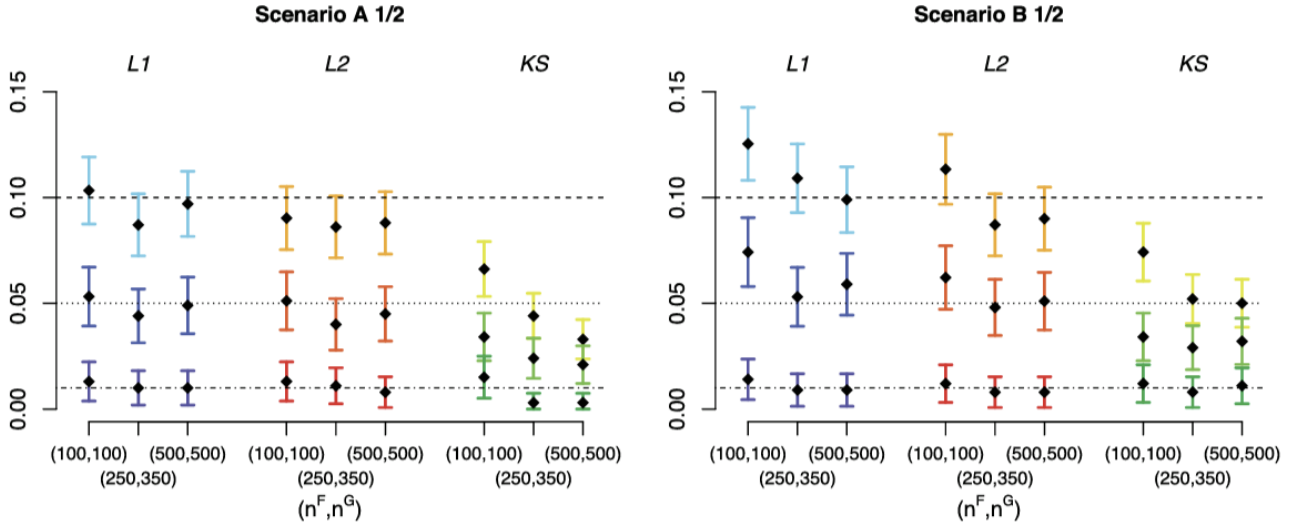


Figure 2: We split the sample in $n_R^F = n_A^F = 1/2n^F$ and $n_R^G = n_A^G = 1/2n^G$. For the L_1 and L_2 statistics, the nominal levels are generally well approximated in both scenarios. This represents the best partition choice in terms of power.

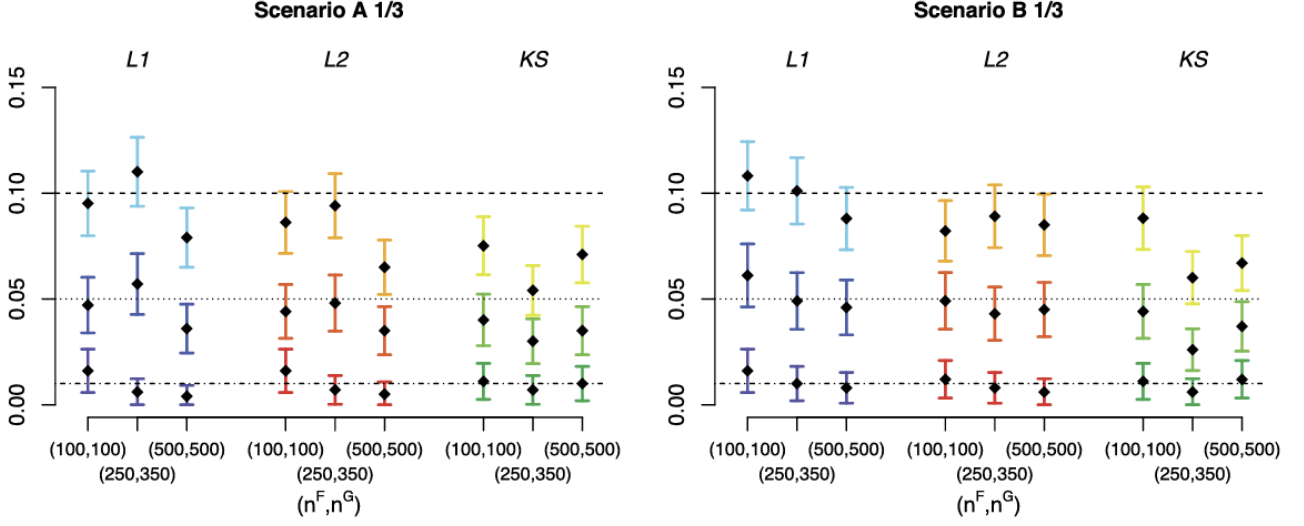


Figure 3: We split the sample in $n_R^F = 1/3n^F$, $n_A^F = 2/3n^F$ (and similarly for the healthy population). In this case, we have less data for the Pooled ROC because the AROC estimation requires more data to improve its accuracy. There is an improvement in the calibration of the KS statistic, while L_1 and L_2 statistics remain stable but suffer from a loss of power.

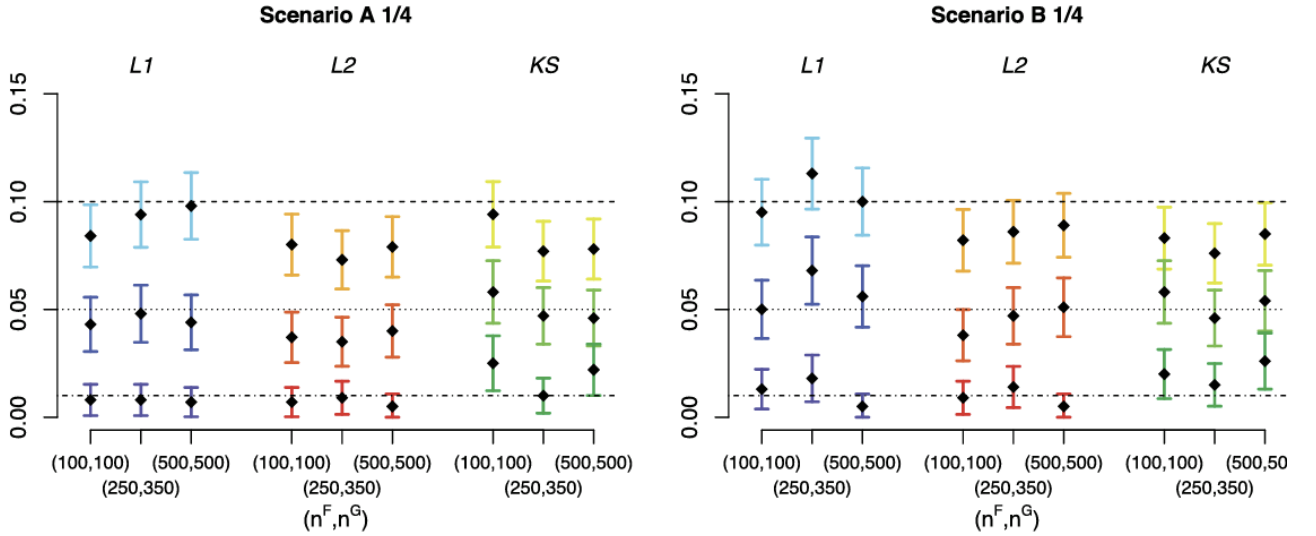


Figure 4: We split the sample in $n_R^F = 1/4n^F$, $n_A^F = 3/4n^F$ (and similarly for the healthy population). We use more data for the AROC since it is more complex to calculate. The calibration of the KS statistic improves mainly with small sample sizes, but the test suffers from a significant loss of power. This is a good choice only with small sample sizes.

Scenarios A and B were selected to analyze the **level** of the test (calibration), as they satisfy the null hypothesis ($ROC(p) = AROC(p)$). Scenarios C and D were employed to analyze the **power** of the test. In general, the power increases with the sample size for all statistics, demonstrating the consistency of the test.

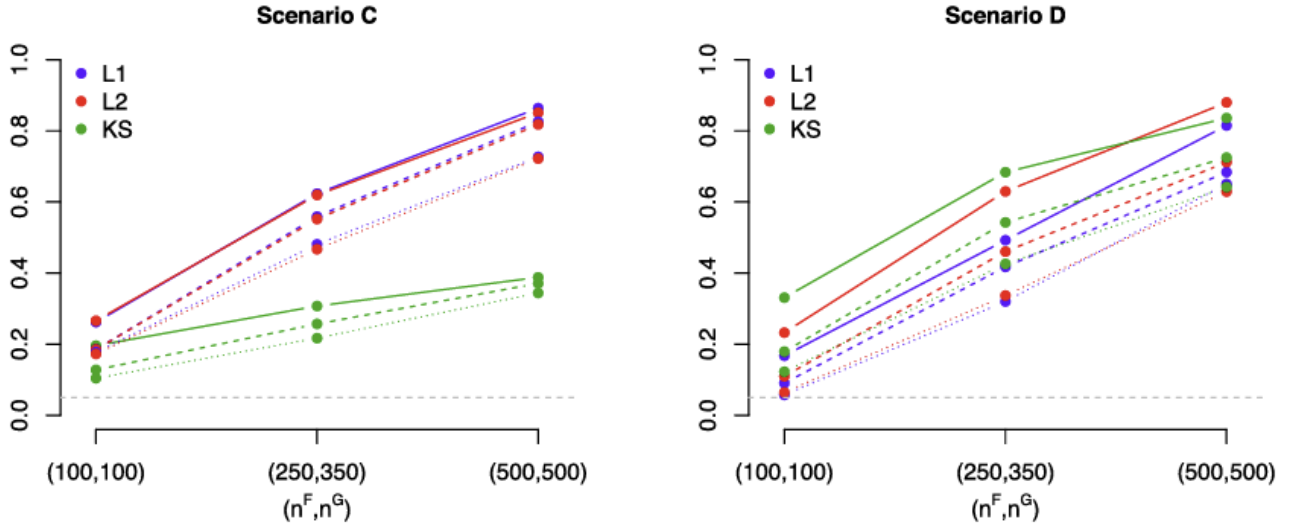


Figure 5: The x -axis represents the sample size, the y -axis represents the power. The solid lines represent the $1/2$ partition, the dashed lines the $1/3$ partition, and the dotted lines the $1/4$ partition. The graph demonstrates that the $1/2$ partition yields the highest power; this is also why it is recommended by the authors for larger sample sizes.

4 Application

A dataset concerning patients suspected of prediabetes, provided by Dr. F. Gude Sampedro (*Unidad de Epidemiología Clínica, Hospital Clínico Universitario de Santiago*), was analyzed.

A patient is considered **prediabetic** if they present a diagnosis of diabetes mellitus or if blood glucose levels are above 100 mg/dL (covariate threshold). The sample consists of 1496 patients, with an unbalanced sample size of 405 considered diseased and 1091 healthy where a binary response variable is used to indicate the presence of prediabetes.

Three diagnostic markers were considered: **GA** (glycated albumin), **a1c1** (haemoglobin), and **-GP22** (glycan peaks), along with the covariate **age**. Notably, higher values of these diagnostic variables are more common in the healthy population rather than in the diseased one.

The main objective is to assess the capability of these three diagnostic variables to correctly diagnose prediabetes while taking into account the covariate age. The study highlights two problems regarding this analysis:

1. The Conditional ROC curve is estimated locally; therefore, the estimations computed on the extreme values of the covariate are less reliable, as there are fewer data points available in those regions.
2. Standard representations often provide no insight into how the covariate is distributed within the healthy and diseased populations.

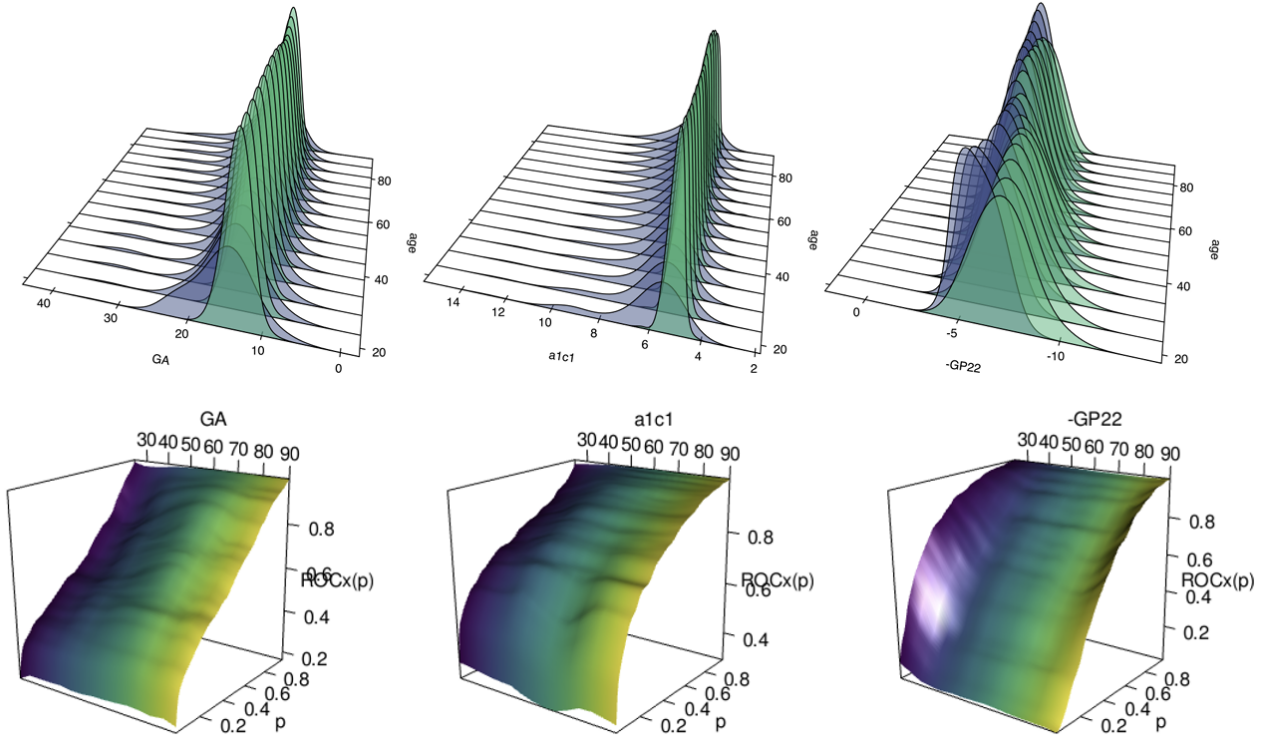


Figure 6: The three graphs on top show the estimated conditional density functions for healthy (blue) and diseased (green) patients, considering age as a covariate. For GA and a1c1, the two curves overlap significantly across all ages. In contrast, for -GP22, the separation is distinct at younger ages, whereas for older ages, the overlap increases (becoming less distinct). The three graphs below display the 3D ROC curves: the x -axis represents specificity, the depth represents age, and the vertical axis represents sensitivity. For GA and a1c1, the surface exhibits a consistent shape across all ages, suggesting the covariate does not affect the accuracy of the marker. Conversely, for -GP22, the surface is arched for younger subjects but flattens for older ones, indicating better discriminatory power for younger patients.

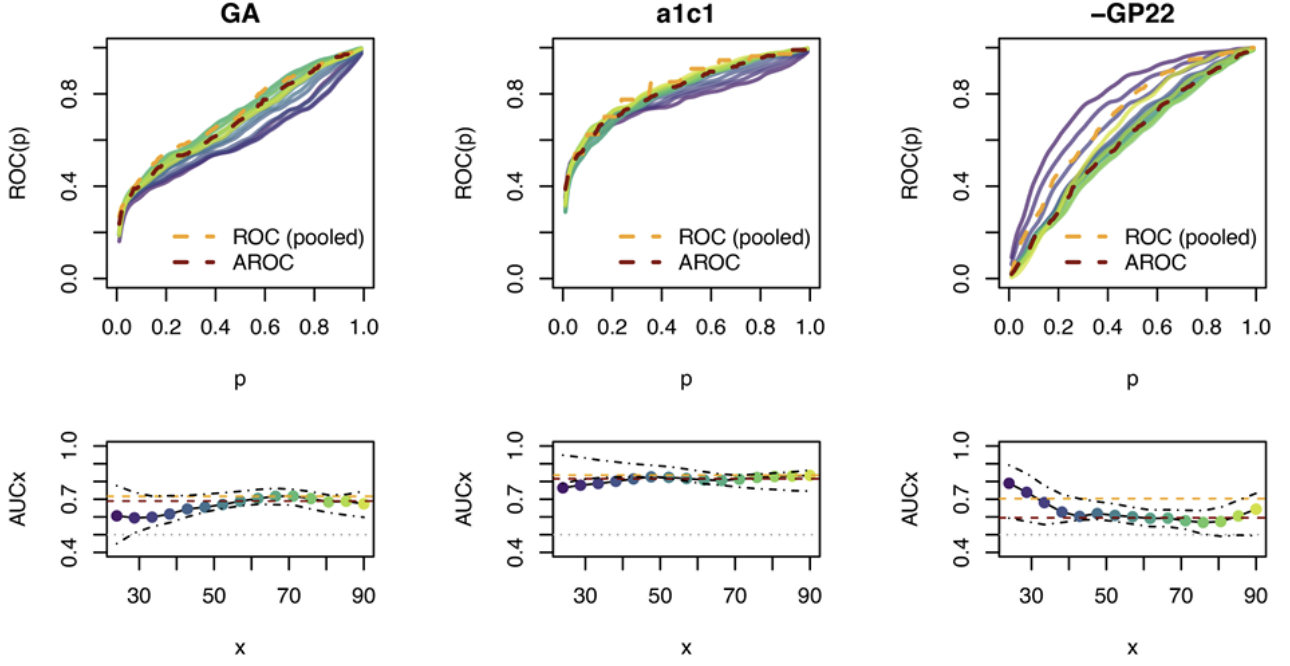


Figure 7: The bottom panels display the Conditional AUC (AUC_x) with age on the x -axis. The colored dotted lines represent the specific AUC for each age, while the straight orange horizontal line represents the AUC of the Pooled ROC. For GA and a1c1, the proximity of these values implies that the covariate effect is negligible. However, for -GP22, the Conditional AUC varies significantly, indicating that the test is more accurate for younger individuals compared to older ones. The top panels represent the ROC curves: for GA and a1c1, the Pooled ROC and AROC overlap, which is confirmed by the flatness of the AUC curve. For -GP22, the Pooled ROC and AROC show a clear separation.

We introduce two hypothesis tests to determine the appropriate ROC curve analysis:

1. **Test 1** assesses whether the diagnostic capability is stable across covariate values.

$$H_0 : \text{ROC}^x(p) = \text{AROC}(p)$$

If we accept H_0 , age does not distort the local performance of the marker.

2. **Test 2** assesses whether the Pooled ROC gives accurate results compared to the AROC.

$$H_0 : \text{AROC}(p) = \text{ROC}(p)$$

If we accept H_0 , it is possible to ignore age and use the Pooled ROC.

We consider a significance level of $\alpha = 0.1$. The results of the tests are summarized in Table 1.

Marker	Test 1 p-value	Test 2 p-value	Conclusion
a1c1	0.853 ($> \alpha$)	0.782 ($> \alpha$)	Use Pooled ROC
-GP22	0.523 ($> \alpha$)	0.001 ($< \alpha$)	Use AROC
GA	0.073 ($< \alpha$)	0.910 ($> \alpha$)	Use Conditional ROC

Regarding **Test 1**, for both -GP22 and a1c1, the p-values (0.523 and 0.853 respectively) imply that we do not observe significant changes in the curves as age varies. Conversely, for

GA, the p-value is 0.073; since $p < 0.1$, we reject the null hypothesis, indicating that age affects the local performance of the test.

Regarding **Test 2**, for GA and a1c1, the p-values (0.910 and 0.782) suggest that the covariate does not affect the curve relative to the Pooled version. However, for -GP22, the p-value (0.001) leads to the rejection of the null hypothesis.

In conclusion:

- For **a1c1**, the best option is the **Pooled ROC**, as we accepted the null hypothesis in both tests.
- For **-GP22**, we use the **AROC** curve, since we accepted H_0 in the first test (stability) but rejected it in the second.
- For **GA**, we employ the **Conditional ROC**, because we rejected the null hypothesis in Test 1, implying that diagnostic accuracy varies with age.

These conclusions partially match our previous hypothesis: while age does not significantly impact a1c1, it plays a crucial role for -GP22 and GA depending on the test criteria used.

5 Personal Application and Validation

We applied the same procedure to other more realistic datasets in order to assess if the model performs effectively with noisy real-world data.

Firstly, we analyzed the **BUPA** dataset, which describes possible liver disorders caused by alcohol, taking into account the number of drinks as the covariate, and (although it was not specified) the health condition as a selector. Ultimately, due to the inconsistent results yielded by the test (present on the R code), we were able to notice that the dataset was poorly structured, indeed the selector was based solely on two random groups and not on the disease status of the patient. This became evident when the outcome was showing that patients with very high value in diagnostic marker were diagnosed as healthy when in reality it should be the opposite.

In order to get some better results we chose another dataset this time verifying the selection criteria prior to any simulation. We employed **Indian Liver Patient Records** (ILPD) from Kaggle. The dataset contains a total of 583 patients, 416 healthy and 167 diseased; and 441 male and 142 female. It should be noted that every patient older than 89 is listed as 90 years old. We selected **Alamine Aminotransferase** (ALT) as a diagnostic marker, because it is an indicator of hepatocellular injury which is a damage to liver cells. On the other hand, **age** was selected as the covariate to verify if it affects the distribution of the marker ALT and the discriminatory power. We ran the simulations to study how the test performs when considering real data.

5.1 Exploratory Data Analysis

Firstly, we conducted an analysis with the aim of studying the dataset composition, behavior and distributions to have a global view of the possible results that we might be getting at the end.

As shown in Figure 8 the distribution of age in both healthy and diseased are not identical but similar. The **median age** for healthy is around 40, meanwhile the one for diseased is 46, so slightly higher. Notably, the diseased group contains more old people compared to the healthy one. This is also further supported by the 3D conditional densities (Figure 9), as a matter

of fact we notice that the ridges for both groups remain **parallel** and maintain a **constant distance**. Noting that the overlap between the two populations is invariant to the covariate.

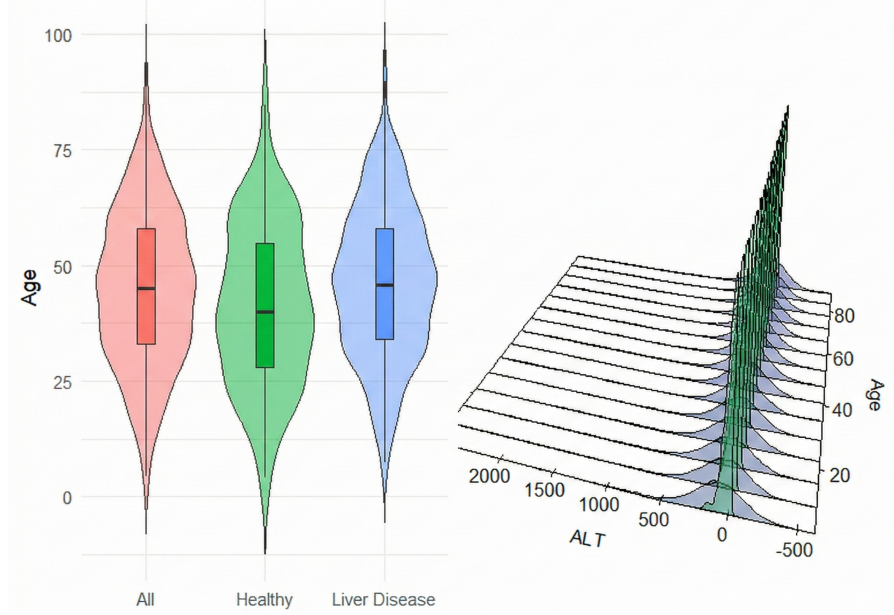


Figure 8: On the left: the violin plot comparing the distribution of age between healthy and diseased, we notice that overall the shapes are quite similar. On the right: the conditional densities of ALT given age, we notice that a general parallelism is present so the distribution of the marker does not change significantly with the covariate.

5.2 Test Results and Critical Discussion

We have now performed the bootstrap-based test to compare Pooled ROC and AROC.

Consistent with simulation results, we employed a **1/2 partition** ($n_R = n_A$) since a balanced data splitting yields the highest power for larger sample sizes. We get three different p-values: 0.79 for L_1 , 0.585 for L_2 and 0.58 for KS . Since the p-values are greater than 0.05 this allows us to accept the null hypothesis.

Now we can study the graphs that we were able to get from the AROC study, but first we have to explain their visual appearance. As a result, the *jagged nature of the ROC curves* is a consequence of the **limited sample size** and the distribution of the dataset. This happens because our dataset is based on real life data compared to the previous one, which was larger and more balanced. Consequently, the two curves are almost overlapping overall, although not perfectly smooth due to the dataset limitations. Note that the gap is a numerical artifact of the sparse data in specific age groups, we can now assume that the age does not significantly improve the diagnostic accuracy of ALT. All these irregularities in our graphs show the challenges of working with **realistic datasets**.

We have plotted the AUC curve and studied it. First, we notice how AUC_x fluctuate around 0.7 for the majority of population (ages 20 to 70). Plotting the Pooled AUC (0.686) and $AAUC$ (0.695) we can explain why they are nearly identical and the AUC_x oscillates between them, indeed as we said before this is a confirmation that adjusting for age does not change the global diagnostic performance.

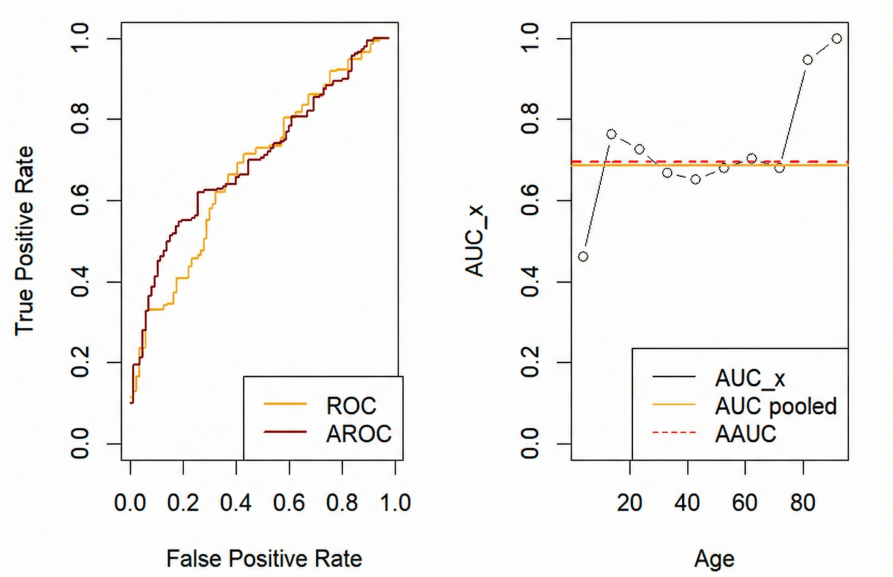


Figure 9: On the left: the estimated Pooled ROC and AROC curves, we notice that the two curves are almost overlapping. On the right: the Conditional AUC plotted against age, we notice that the oscillation in the curve proves that the diagnostic capability is stable across different ages.

6 Discussion

In this report, we have analyzed the relationship between ROC curves and covariates. We discussed the relationships between the Pooled ROC, the Covariate-Adjusted ROC (AROC), and the Conditional ROC curves, proposing a nonparametric test to assess the validity of using the simpler Pooled ROC curve. Beyond the theoretical framework already provided, the application to the ILPD dataset allowed us to observe these behaviors in a real-world context, highlighting the stability of the Pooled ROC when the covariate effect is negligible.

Although we have only focused on unidimensional covariates, the same discussion applies to **multidimensional covariates**. However, this extension requires introducing an adequate estimator for the AROC in higher dimensions and adapting the resampling step in the bootstrap algorithm accordingly.

It is worth noting that most existing tests for comparing ROC curves rely on comparing the areas under the curves (AUC). A limitation of this approach is that equal AUCs do not necessarily imply identical ROC curves. The methodology applied here compares the entire curves, avoiding this issue.

Finally, being able to correctly identify whether to use the Pooled, Conditional, or AROC curve is crucial for practical applications, such as determining optimal cut-off points for diagnostic tests. If the null hypothesis holds, the Pooled ROC allows for covariates independent threshold; otherwise, covariate-specific thresholds should be considered to ensure diagnostic accuracy.

Note on Methodology

This paper reproduces the simulation settings and adopts the testing procedure described by Fanjul-Hevia et al. [1]. The analysis performed on the BUPA[2] and ILPD[3] datasets constitutes the original application of this study.

References

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